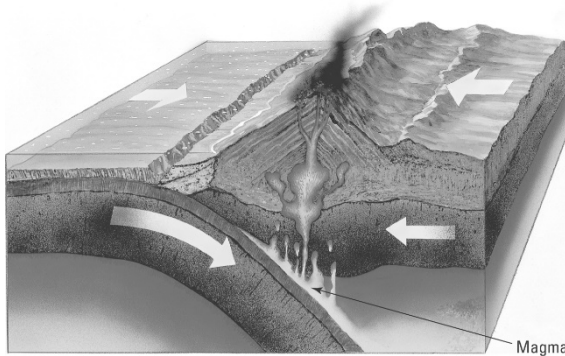


Volcanic activity

Answers

Steep-sided volcanoes

1. The diagram shows a steep-sided volcano forming as a result of tectonic activity.



- (a) Identify the type of plate boundary that leads to the formation of the volcano.

Subduction zone

- (b) Explain why the oceanic crust melts to form magma.

Frictional heating

- (c) Identify a location in the southern hemisphere where such an event occurs.

West coast of South America

Eruption

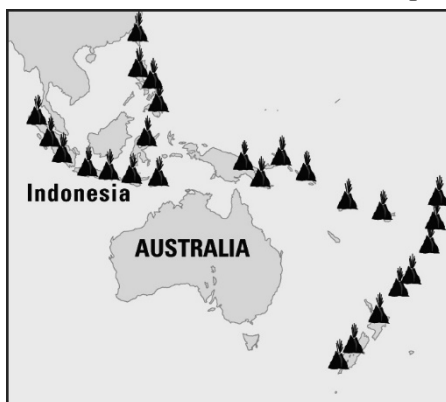
2. The following events describe the eruption of a steep-sided volcano. The events are jumbled. Rearrange the letters A–F to arrange the events in the correct sequence.

- A. Magma rises towards the surface.
- B. Lava is blasted into the air.
- C. The expanding gases cause an explosion that rips open the vent.
- D. Ash, cinders and dust rain down onto the ground.
- E. Low-density magma rises under pressure into the magma chamber.
- F. Dissolved gases expand as the magma rises.

Answer: E, A, F, C, B, D

Our volcanic neighbours

3. The diagram below shows the region north and east of Australia. Volcanoes are marked on the map.



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- (a) Explain why the volcanoes are located in these places. The volcanoes are present along the boundaries of plates. These plate boundaries are subduction zones.

- (b) Explain why there are no active volcanoes in Australia. Australia does not lie near the edge of a plate (the Indo-Australian plate). We have a small number of earthquakes. However, since the continent is a long way from a subduction zone or a slip-strike zone, volcanoes do not form.